

ASCII MASTER FLOW – PANEL 1/12: Fission-to-grid chain

FISSION -> heavy nucleus splits and releases heat plus neutrons

COOLANT -> removes core heat

STEAM SYSTEM -> converts heat into working-fluid energy

TURBINE-GENERATOR -> produces electricity

RULE -> reactor capacity is not electricity generated

ASCII MASTER FLOW – PANEL 2/12: Criticality milestone ladder

SUBCRITICAL -> chain reaction declines

FIRST CRITICALITY -> controlled self-sustaining reaction begins

POWER ASCENSION -> tested increase toward rated conditions

GRID SYNCHRONISATION -> generator connects to grid

COMMERCIAL OPERATION -> later declared operating status

ASCII MASTER FLOW – PANEL 3/12: Core-control function map

CONTROL RODS -> absorb neutrons and regulate reactivity

MODERATOR -> slows neutrons in a thermal reactor

COOLANT -> transfers heat out of the core

CONTAINMENT -> barrier against radioactive release

RULE -> four functions, four different answers

ASCII MASTER FLOW – PANEL 4/12: Thermal-fast comparison

THERMAL REACTOR -> moderated slower-neutron spectrum
FAST REACTOR -> no moderator and fast-neutron spectrum
PHWR -> heavy-water moderated Stage-1 workhorse
FBR -> breeding objective with fertile blanket
RULE -> coolant alone does not define the spectrum

ASCII MASTER FLOW – PANEL 5/12: Fissile-fertile conversion map

U-235 / PU-239 / U-233 -> fissile materials

U-238 -> neutron capture -> PU-239 pathway

TH-232 -> neutron capture and decay -> U-233 pathway

FERTILE MATERIAL -> needs irradiation and conversion

RULE -> resource abundance is not ready reactor fuel

ASCII MASTER FLOW – PANEL 6/12: Three-stage programme rail

STAGE 1 -> PHWR, natural uranium and plutonium-bearing spent fuel

REPROCESSING -> separates material for the breeder bridge

STAGE 2 -> fast breeder and fissile-inventory multiplication

THORIUM IRRADIATION -> prepares U-233 route

STAGE 3 -> long-term thorium-U-233 utilisation horizon

ASCII MASTER FLOW – PANEL 7/12: PFBR status card

DESIGN -> IGCAR fast-reactor and sodium expertise

BUILD / COMMISSION -> BHAVINI project responsibility

RATING -> 500 MWe design value in fetched DAE text

6 APRIL 2026 -> first criticality achieved

NOT ESTABLISHED -> grid synchronisation or commercial operation

ASCII MASTER FLOW – PANEL 8/12: Closed fuel-cycle loop

REACTOR IRRADIATION -> changes isotopic composition

COOLING / HANDLING -> manages heat and radiation

REPROCESSING -> separates reusable fissile material

REMOTE FABRICATION -> prepares new fuel

REUSE / WASTE -> closes material accounting loop

ASCII MASTER FLOW – PANEL 9/12: Institutional separation

DAE -> governmental policy and strategic umbrella

BARC -> reactor and fuel-cycle R&D

NPCIL -> commercial civilian plant developer and operator

AERB -> safety review and regulatory oversight

RULE -> promoter, laboratory, operator and regulator differ

ASCII MASTER FLOW – PANEL 10/12: Fuel-and-project chain

UCIL -> uranium mining and milling

NFC -> fuel and reactor-component fabrication

IREL -> monazite, thorium and rare-earth processing link

IGCAR -> fast-reactor design and sodium R&D

BHAVINI -> PFBR project execution and operation

MUST REMEMBER: Fission splits heavy nuclei and is controlled through fuel, moderator,...

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ASCII MASTER FLOW – PANEL 11/12: Reactor taxonomy

PHWR -> heavy water and natural-uranium Stage-1 role

PWR / BWR -> light-water systems with different steam cycles

FBR -> fast spectrum and breeding objective

AHWR / SMALL-REACTOR LINES -> design or development status

RULE -> type and deployment status must be stated together

CLOSE DISTINCTION: Fissile is not fertile, enrichment is not reprocessing, heavy water is...

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ASCII MASTER FLOW – PANEL 12/12: Nuclear answer spine

DEFINE -> fission, criticality, fissile, fertile and fuel cycle

TRACE -> Stage 1, plutonium bridge, Stage 2 and thorium horizon

MAP -> DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain

SEPARATE -> capacity criticality grid generation and legal status

VERIFY -> dated fleet fuel reactor law cost and schedule evidence

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Trace PHWR uranium, fast-breeder...

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